



Credit Risk Analysis Report

1. Dataset Preprocessing Steps

- **Dataset:** The dataset (`cs-training.csv`) was read using pandas with the index column specified.
- **Missing Value Treatment:** Missing values were filled with column medians to prevent bias and retain distribution.
- **Column Renaming:** All column names were renamed for readability and easier reference.
- **Feature Engineering:**
 - `IncomePerDebt` = `MonthlyIncome` / (`DebtRatio` + 1): captures the relationship between income and debt burden.
 - `TotalPastDue` = Sum of all past due indicators: gives a consolidated view of a customer's delinquency history.
- **Feature Scaling:** All features were standardized using `StandardScaler` to ensure model stability and performance.
- **Class Imbalance Handling:** SMOTE (Synthetic Minority Over-sampling Technique) was used to balance the dataset due to imbalanced classes in the target variable (`SeriousDlqin2yrs`).

2. Model Selection and Rationale

Three models were chosen for evaluation:

Model	Rationale
Random Forest	Robust, handles feature importance well, performs well on tabular data.
Gradient Boosting	Builds trees sequentially to correct previous errors; good for handling class imbalance.
XGBoost	Optimized boosting algorithm known for its efficiency and high performance in structured datasets.

3. Challenges Faced and Solutions

Challenge	Solution
Missing data in income	Filled using median to avoid distortion of distribution.
Highly imbalanced target class	Addressed using SMOTE for oversampling the minority class.
Model overfitting risk	Split data into train-test and used scaled features to minimize variance.

4. Results with Visualizations and Interpretations

Random Forest Model

[[26665 1260]

[2018 26047]]

	precision	recall	f1-score	support
0	0.93	0.95	0.94	27925
1	0.95	0.93	0.94	28065
accuracy			0.94	55990
macro avg	0.94	0.94	0.94	55990
weighted avg	0.94	0.94	0.94	55990

ROC AUC Score: 0.9415

Gradient Boosting Model

[[24666 3259]

[4161 23904]]

	precision	recall	f1-score	support
0	0.86	0.88	0.87	27925
1	0.88	0.85	0.87	28065
accuracy			0.87	55990
macro avg	0.87	0.87	0.87	55990
weighted avg	0.87	0.87	0.87	55990

ROC AUC Score: 0.8675

XGBoost Model

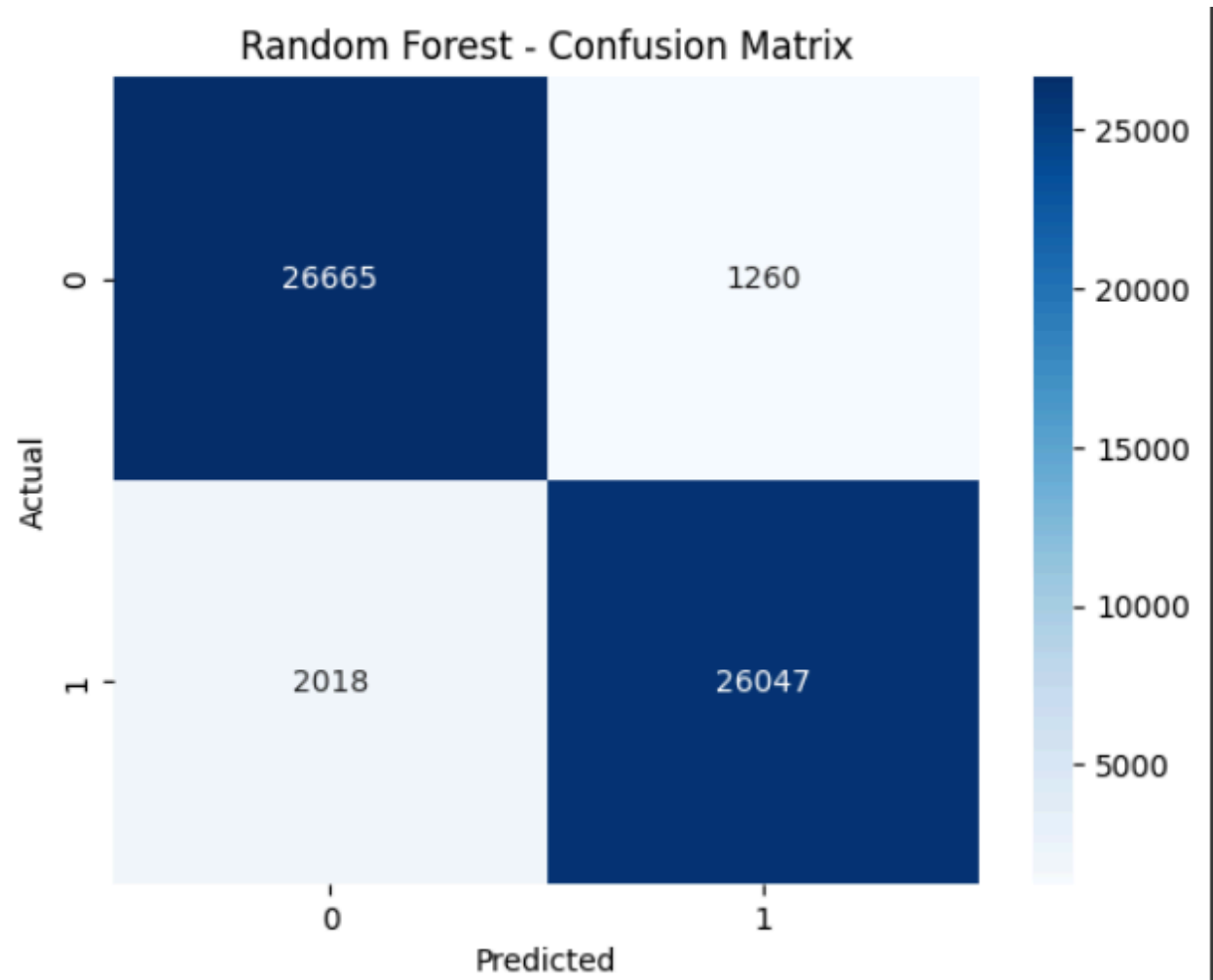
[[26943 982]

[2692 25373]]

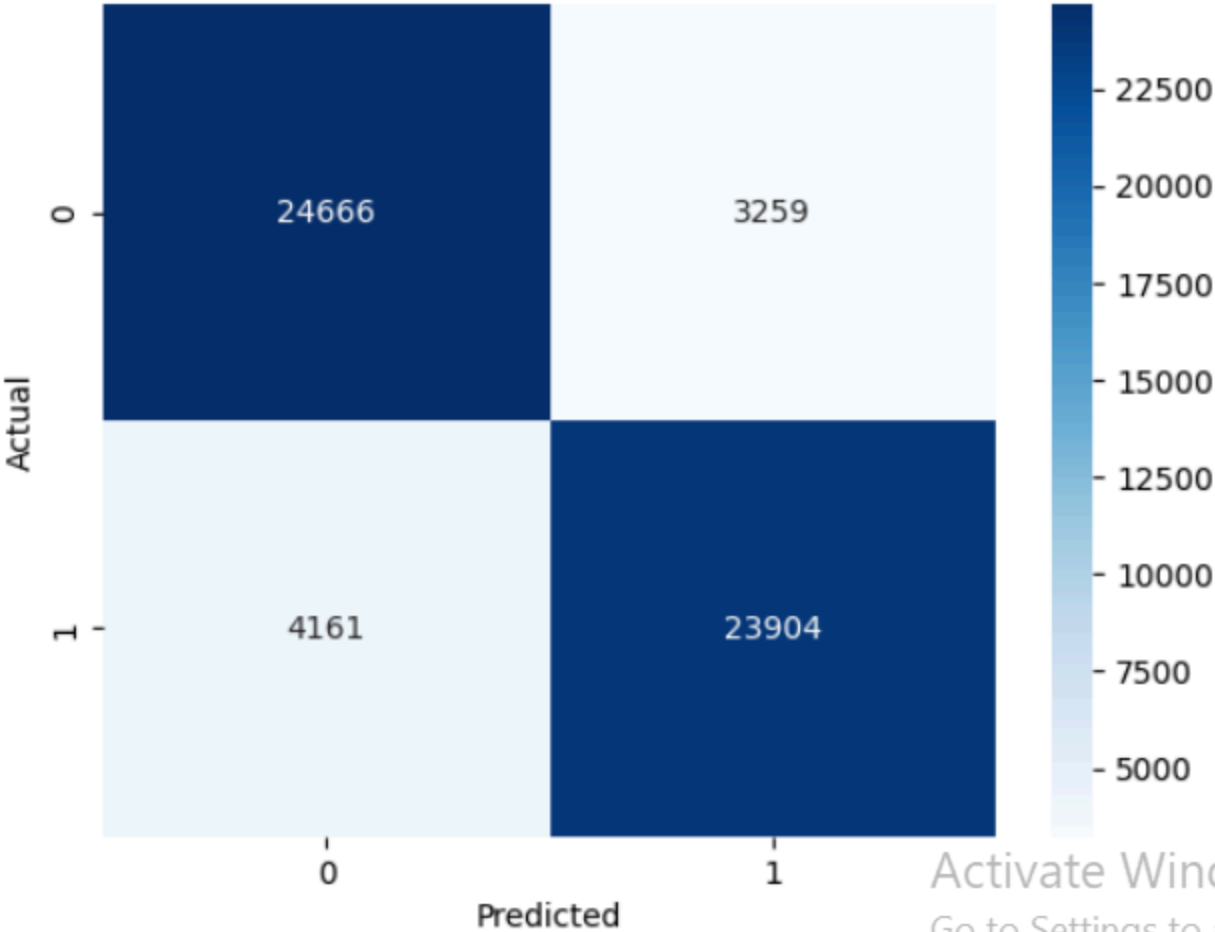
	precision	recall	f1-score	support
0	0.91	0.96	0.94	27925
1	0.96	0.90	0.93	28065
accuracy			0.93	55990
macro avg	0.94	0.93	0.93	55990
weighted avg	0.94	0.93	0.93	55990

ROC AUC Score: 0.9345

Confusion Matrix



Gradient Boosting - Confusion Matrix



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